Business Requirement Document

Project	Lending Default Risk Analysis
Version	1.0
Author	Archith
Data source	Home Credit Default Risk (Kaggle)
Tools	SQL (data preparation and cleaning), Power BI (modelling, DAX, report)
Deliverable	4-page interactive Power BI report

Executive Summary

Digital Lending Credit Decision Analytics is a portfolio analytics project built around the Kaggle Home Credit Default Risk dataset. It brings applicant-level attributes together with summarised historical credit and repayment information so users can explore the lending portfolio and understand how observed default rates vary across applicant segments.

MySQL is used for data staging, profiling, transformation, and applicant-level aggregation; Power BI is used for interactive reporting. The planned report pages are Lending Portfolio, Applicant Risk Patterns, Credit & Repayment, and Decision Support. The report presents portfolio metrics, segment comparisons, historical behaviour indicators, and transparent what-if analysis.

Business Problems

Business Objectives

ID	Objective
O1	Build one trusted dataset
O2	Show default risk by segment
O3	Test loan-size rules against a baseline
O4	Rank the drivers of default
O5	Give actionable recommendations
O6	Keep the report explainable and interactive

Scope

The project covers descriptive and diagnostic analysis of default risk on existing borrowers; it does not build or deploy a predictive model.

In scope

- Preparing, aggregating and cleaning the application, bureau, previous application and instalment tables in SQL
- A Power BI data model, DAX measures and a 4-page interactive report
- A policy simulator that back-tests a credit-to-income cap against random rejection
- Segment analysis by credit history, repayment behaviour, loan size and age
- A Findings and Recommendations page with one view per topic: Late Payments, Historical Credit Behaviour, Borrowing Volume and Age Group

Out of scope

- A production credit scoring model or automated approve/decline decisions
- Analysis of applicants who were actually rejected (the data holds only approved loans)
- Trends over calendar time (the dataset has relative day counts, not dates)
- Live data refresh or integration with a lender's systems
- Pricing, interest income and profitability analysis

Success Criteria

- Every page counts the same 308K applicants and totals reconcile
- Default rate shown with applicant volume for every band and segment
- Policy simulator compares each cap with random rejection (baseline 11.39)
- Credit history, repayment, loan size and age compared on the same scale
- Each recommendation carries a finding, an action and a success metric
- Slicers, simulator and recommendation buttons work on every page

Business Rules

ID	Business rule
BR-01	TARGET = 1 marks an applicant with payment difficulties, treated as default; TARGET = 0 means no recorded default. This applies to the supplied application training data only.
BR-02	Default Rate = Total Defaults ÷ Total Applications in the current filter context. If there are no applications, show a no-data state, not 0%.

BR-03	Missing history (no bureau record, no prior application) is never read as good or bad behaviour. It is shown as its own labelled group, such as “No bureau record”.
BR-04	Segment rates are read together with applicant counts. Segments below 500 applicants are not named as the highest risk or treated as definitive rankings.
BR-05	A difference in observed default rates shows an association, not that the attribute caused default.
BR-06	The policy simulator describes historical records above a chosen credit-to-income cap. Labels must say “would have”, and must not claim defaults were definitively avoided.
BR-07	Age and similar attributes are used only to explore patterns. Any real lending use needs fairness, legal and compliance review.

Reference Documents

Data set : Home Credit Default Risk

Project’s GitHub Repository: GitHub - ArchithJaiswal/Lending-Default-Risk-Analysis